Cutaneous T-cell lymphoma

Mycosis fungoides · Sézary syndrome · primary cutaneous CD30+ LPDs — **skin-directed therapy first, systemic therapy by compartment (skin vs blood), allo-HCT for selected patients**

Compiled from the Stanford Oncology Handbook 2025, Mycosis Fungoides/Sézary Syndrome (Chou L, Khodadoust M) · NCCN Primary Cutaneous Lymphomas · primary trial reports and FDA actions verified against PubMed, September 2026.

WHAT YOU ARE DEALING WITH

Entity	Defining features
Mycosis fungoides	Most common CTCL. Patch → plaque → tumor, often in sun-protected "bathing-trunk" skin. Epidermotropic CD4+ cells, **CD7 loss**, Pautrier microabscesses. Skin-resident memory T-cell origin
Sézary syndrome *<5% of CTCL*	**All three:** erythroderma (≥80% BSA) + **B2 blood** + **a blood clone matching the skin**. Central-memory origin — a leukemic disease, not simply late MF
Adverse MF variants	**Folliculotropic** (independent predictor of progression and death) · **large-cell transformation** (>25% large cells; higher progression risk) · tumor stage, raised LDH
Primary cutaneous CD30+ LPDs *2nd most common CTCL*	**Lymphomatoid papulosis** and **primary cutaneous ALCL** — indolent, excellent prognosis despite anaplastic cytology. Distinguish from systemic ALCL spreading to skin (ALK+ = systemic until proven otherwise)

A **TCR clone alone is not diagnostic** — clonal T cells occur in benign dermatoses. Early MF often needs several biopsies, ideally taken off topical steroids.

WORKUP

- **Skin biopsy** (multiple sites) with IHC incl. **CD30** and TCR gene rearrangement; **measure BSA** (patient palm + fingers ≈ 1%) and mSWAT
- **Blood flow cytometry** (CD4:CD8, CD4+CD7−, CD4+CD26− absolute counts) + **blood TCR clone** — needed to assign B stage
- **Imaging: PET/CT or CT for T2b and above**, palpable nodes, or B1–B2 blood. Stage IA with no nodes needs no imaging
- **Excise an abnormal node** — nodal histology (N1–N3) changes the stage; marrow only if unexplained cytopenias
- LDH, CBC, CMP; **HTLV-1 serology** if erythroderma or from an endemic area (ATLL mimics Sézary)

ISCL/EORTC TNMB (OLSEN, BLOOD 2022)

T1 · T2	Patches/plaques **<10%** · **≥10% BSA** (a = patch only, b = plaque)
T3 · T4	≥1 **tumor ≥1 cm** · **erythroderma ≥80% BSA**
N1 · N2 · N3	Abnormal node, histology Dutch grade 1 / 2 / 3–4 (NCI LN 0–2 / 3 / 4)
M1	**Biopsy-proven** visceral disease
B0	≤5% Sézary cells, or <250/μL, or <15% CD4+CD26− / CD4+CD7−
B1	More than B0 but not B2
B2	**≥1,000/μL** Sézary cells, or CD4+CD26− / CD4+CD7− / other aberrant cells by flow — **plus a blood clone matching the skin**

STAGE GROUPS

Stage	T	N	M	B
IA · IB	1 · 2	0	0	0–1
IIA	1–2	1–2	0	0–1
IIB tumor	3	0–2	0	0–1
IIIA · IIIB erythrodermic	4	0–2	0	0 · 1
IVA1 = Sézary	any	0–2	0	**2**
IVA2 · IVB	any	3 · any	0 · 1	any

Stage IA: 5-yr OS 94%, median survival ≥20 yr — most die of something else. Stage III–IV: median survival ~5 yr, and over half die of disease (Agar, JCO 2010; NCI PDQ).

TREATMENT PATHWAY

Early MF — IA to IIA

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Skin-directed therapy
IA: topical steroid, **mechlorethamine gel**, bexarotene gel, imiquimod, local RT
IB–IIA: **nbUVB** (patches) or **PUVA** (plaques)
not curative — **do not escalate for cosmetics**

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Refractory or rapidly relapsing
add a low-intensity systemic: bexarotene, IFN, low-dose MTX; **TSEBT 12 Gy** for widespread

Tumor stage IIB · LCT

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Limited vs generalized
limited tumors: **local RT 8–12 Gy** + skin therapy
generalized: **TSEBT** or systemic
CD30+ or LCT: brentuximab (ALCANZA)
others: HDACi, mogamulizumab, gemcitabine, liposomal doxorubicin, pralatrexate

Erythrodermic III · Sézary IVA1

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Treat the blood
mogamulizumab (best blood responses)
ECP ± IFN or bexarotene — lower blood burden
romidepsin, low-dose alemtuzumab, pembrolizumab
+ skin care, anti-staph, antipruritics

Advanced, refractory, fit

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Allogeneic HCT
only curative option; **go in remission**
CUTALLO: PFS 9.0 vs 3.0 mo (HR 0.38)
TSEBT often used to debulk skin pre-transplant

CD30+ LPD

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Lymphomatoid papulosis · pc-ALCL
LyP: observe / topicals / **low-dose MTX**
pc-ALCL solitary: **excision or RT**
multifocal / relapsing: MTX or **brentuximab**
multi-agent chemo is over-treatment

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SKIN-DIRECTED THERAPY

Therapy	How and what to know
Topical corticosteroids	High-potency (clobetasol) BID for T1 patches — high response rates in limited disease. Atrophy, striae, HPA suppression with large BSA
Mechlorethamine gel 0.016%	Daily thin layer to lesions; gel non-inferior to compounded ointment (CAILS ORR **59% vs 48%**), faster response. **Contact dermatitis ~20%** — reduce frequency, add topical steroid. No systemic absorption
Bexarotene gel 1% · imiquimod	Retinoid gel titrated to BID–QID for limited lesions; irritation. Imiquimod for thick single plaques
nbUVB · PUVA	**nbUVB for patches**, **PUVA for thicker plaques** (deeper penetration); 2–3×/wk to clearance, then taper. **Cumulative risk of skin cancer** (PUVA: SCC, melanoma); eye protection after psoralen
Local RT	**8–12 Gy** per lesion for palliation; **24–30 Gy for unilesional disease** (curative intent). MF is exquisitely radiosensitive
TSEBT	**12 Gy low-dose** (1 Gy ×12): ORR **88%**, CR 27%, median clinical benefit ~71 wk, **repeatable**. 30–36 Gy gives deeper CR but more toxicity and only once or twice per lifetime

Maximise skin-directed therapy and supportive care first — it lengthens time to the next systemic. Check that a systemic agent is compatible with phototherapy/TSEBT before combining.

PRIMARY CUTANEOUS CD30+ LPDS

- **Lymphomatoid papulosis:** crops of papules that **self-heal in weeks**. No therapy changes survival — observe, topical steroids, phototherapy, or **low-dose weekly MTX** for scarring or frequent crops. **Lifelong follow-up**: associated second lymphomas (MF, ALCL, Hodgkin)
- **Primary cutaneous ALCL:** solitary/localized nodules, >75% CD30+, ALK–. **Excision or RT**; multifocal → MTX or brentuximab. Staging CT/PET to exclude systemic ALCL

SYSTEMIC THERAPY — DOSING, EVIDENCE, MONITORING

Agent	Dose	Evidence · what to watch
Bexarotene *RXR, oral*	**300 mg/m²/day** with food; often started lower and titrated	ORR 45% in refractory IIB–IVB. **Hypertriglyceridemia** (pancreatitis) — start **fenofibrate** first, **avoid gemfibrozil** (raises bexarotene levels). **Central hypothyroidism: TSH falls — dose levothyroxine to free T4**. Teratogen
Interferon-α	3–6 MU SC daily or TIW (pegylated IFN now used — Intron A off market)	Flu-like symptoms, depression, cytopenias, thyroid. Pairs well with ECP, bexarotene, PUVA
Low-dose methotrexate	10–25 mg PO/SC weekly + folate	Workhorse for MF and **LyP**. LFTs, CBC; switch to pralatrexate 15 mg/m² if more is needed
Vorinostat *HDACi*	400 mg PO daily	ORR **30%**, pruritus relief 32%. Diarrhea, fatigue, thrombocytopenia, **PE 5%**
Romidepsin *HDACi*	14 mg/m² IV d1, 8, 15 q28d	ORR **34%**, DOR 15 mo, **pruritus better in 43%**. Nausea; **QTc — replete K/Mg**. CTCL approval intact
Brentuximab vedotin	1.8 mg/kg q21d, up to 16 cycles	ALCANZA (CD30 ≥10%): ORR4 **56% vs 13%**, PFS 16.7 vs 3.5 mo. **Neuropathy 67%**. Best for plaques/tumors, **LCT**, pc-ALCL
Mogamulizumab *anti-CCR4*	1 mg/kg IV d1, 8, 15, 22 of cycle 1, then d1, 15 q28d	MAVORIC: PFS **7.7 vs 3.1 mo**; ORR 28% vs 5%, **Sézary 37% vs 2%** — blood clears best. **Drug eruption in ~24%**, median onset **119 days**, mimics progression — **biopsy + clonality before calling relapse**. Deplete Tregs: **raises GVHD if given shortly before allo-HCT**
Denileukin diftitox-cxdl *IL-2–toxin*	9 µg/kg/day IV d1–5 q21d, up to 8 cycles	**FDA Aug 2024**, stage I–III after ≥1 systemic: ORR **36%**, CR 9%. **Capillary leak 20%** (boxed) — weight, albumin, edema; infusion reactions, hepatotoxicity
ECP	2 consecutive days q2–4 wk	Erythrodermic / Sézary with modest blood burden; nearly non-immunosuppressive. Combine with IFN or bexarotene
Pembrolizumab *off-label*	200 mg q3wk	ORR **38%** in advanced MF/SS; **Sézary flare (erythroderma, itch) in 53%** — not progression
Alemtuzumab *low-dose, historical*	3 mg d1, then 10 mg SC TIW, max 12 wk	**Sézary / blood only** (skips skin-resident MF cells). **CMV PCR q2–3 wk** + acyclovir + TMP-SMX. Off the US market — access program
Gemcitabine · liposomal doxorubicin · pralatrexate	Gem 1,000 mg/m² d1, 8, 15 q28d · PLD 10–30 mg/m² q2–4wk · pralatrexate 15 mg/m² d1, 8, 15 q28d	For tumor stage and LCT. **Pralatrexate at half the PTCL dose**, with B12/folate

Lacutamab (anti-KIR3DL2) has breakthrough designation for Sézary but is **not approved**; phase 3 TELLOMAK-3 starting 2026. Multi-agent chemotherapy (CHOP, ICE, GDP) gives short responses and heavy infection risk — reserve for bridging.

ITCH, SKIN BARRIER AND INFECTION

- **S. aureus colonizes ~50% of erythrodermic patients** and drives flares: in one series, eradication (oral antibiotics + intranasal mupirocin) cleared carriage in >85% and **improved the skin in 58%**. **MRSA is common** — TMP-SMX or doxycycline empirically
- **Dilute bleach baths**, emollients, wet wraps; **culture before calling progression** — infection and drug eruption both mimic it
- **Bacteremia from skin portals** is a major risk in erythrodermic disease — avoid lines through involved skin when possible
- **Pruritus**: often the dominant symptom; antihistamines rarely enough. Gabapentin, mirtazapine, aprepitant; **disease control is the best antipruritic** (romidepsin 43%, vorinostat 32% itch relief)
- **Lifelong skin-cancer surveillance** after PUVA, TSEBT and mechlorethamine

KEY TRIALS

MAVORIC	Mogamulizumab vs vorinostat — PFS 7.7 vs 3.1 mo, HR 0.53; Sézary ORR 37% vs 2%
ALCANZA	BV vs MTX/bexarotene, CD30+ MF or pc-ALCL — ORR4 56% vs 13%
CUTALLO	Allo-HCT vs none (donor availability) — PFS 9.0 vs 3.0 mo, HR 0.38
Study 302	Denileukin diftitox-cxdl — ORR 36%, capillary leak 20%
CITN-10	Pembrolizumab — ORR 38%, durable
Lessin 2013	Mechlorethamine gel vs ointment — non-inferior, faster